HACKTOBERFEST 26

LIVESTREAM · OCTOBER 22, 2026

Making Open Models Smarter with Tools, MCP, and Skills

Build an agent on a local open model, then connect it to the web





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YOUR HOSTS

Two Developer Advocates at SerpApi



Tomás Murúa

DEVELOPER ADVOCATE

Built the local LLM app in today's demo. Search relevance, Elasticsearch and RAG.



Adarsh Divakaran

PYTHON DEVELOPER ADVOCATE

Python and AI agents. Works on SerpApi's MCP server. Microsoft MVP.

01

A Local LLM with Real-Time Web Data

Tool calling, context windows and output formats

THE PROBLEM

Every Model Stops Learning at Its Cutoff



Muse Glimmer

CUTOFF · 2024

"Tim Cook has been Apple's CEO since 2011 and is still in the role as of my 2024 knowledge cutoff."

Bonsai 27B

CUTOFF · 2024

"Based on all available information through 2024, Tim Cook remains the CEO of Apple."

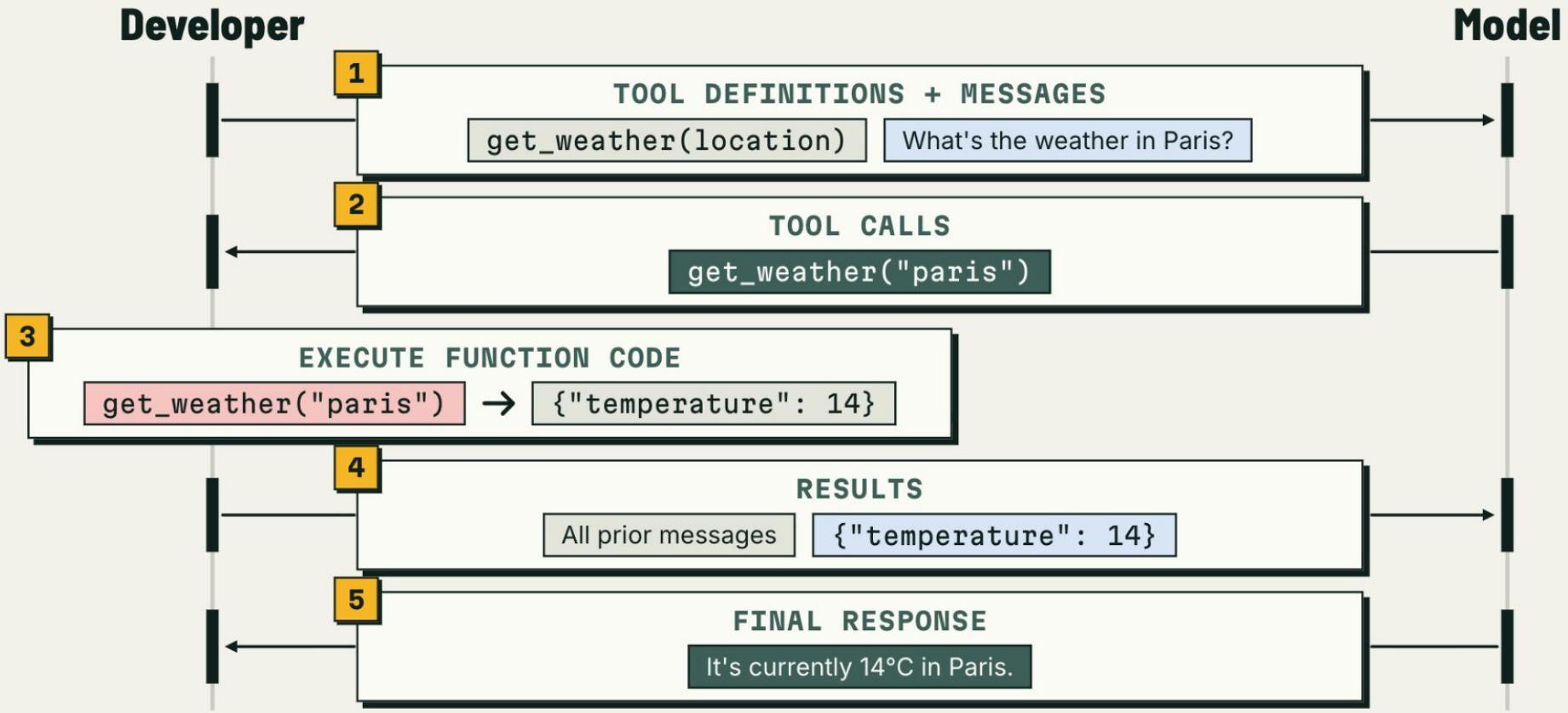
Gemma 4 12B QAT

CUTOFF · JANUARY 2025

"I rely on the information I was trained on, which has a knowledge cutoff of January 2025. As of my last update, Tim Cook remains the CEO of Apple."

TOOL CALLING

The Model Proposes, Your Code Executes



TOOL CALLING · EXAMPLE

The Same Five Steps in Python

```
from openai import OpenAI
import json

client = OpenAI()
tools = [{"type": "function", "name": "get_horoscope",
          "description": "Get today's horoscope for an astrological sign.",
          "parameters": {"type": "object", "properties": {"sign": {"type": "string"}},
                         "required": ["sign"]}}]

def get_horoscope(sign):
    return f"{sign}: Next Tuesday you will befriend a baby otter."

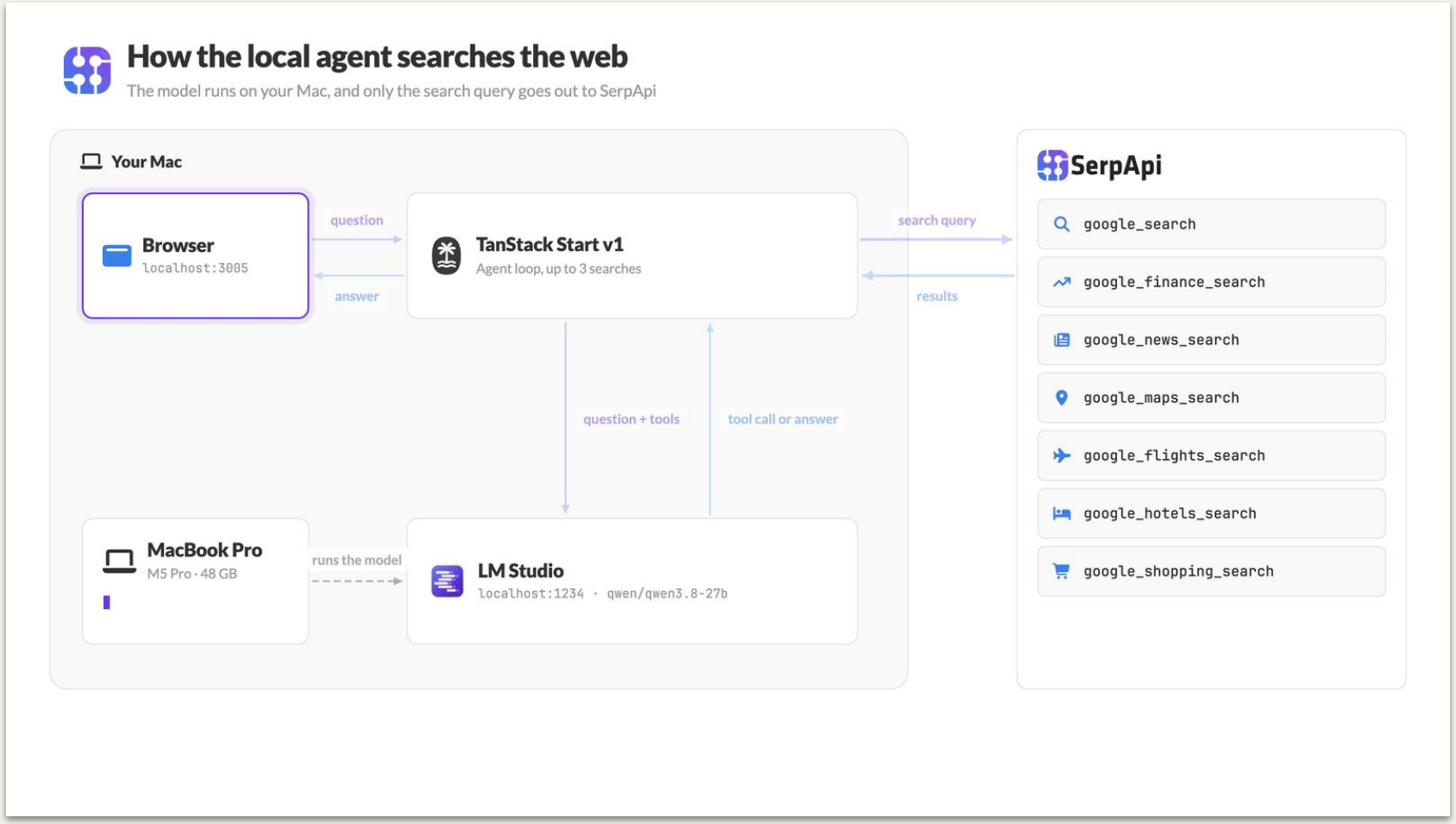
input_list = [{"role": "user", "content": "What is my horoscope? I am an Aquarius."}]
1 response = client.responses.create(model="gpt-6-astra", tools=tools, input=input_list)
input_list += response.output

2 for item in response.output:
    if item.type == "function_call":
3       result = get_horoscope(json.loads(item.arguments)["sign"])
        input_list.append({"type": "function_call_output",
                           "call_id": item.call_id, "output": result})

4 response = client.responses.create(model="gpt-6-astra", tools=tools, input=input_list)
5 print(response.output_text)
```

THE APP

How the App Searches the Web



THE CONTEXT WINDOW

Big Tool Results Fill the Context Window

The context window is everything the model reads in one call: the system prompt, the tools, the chat and every search result.

JSON, RAW

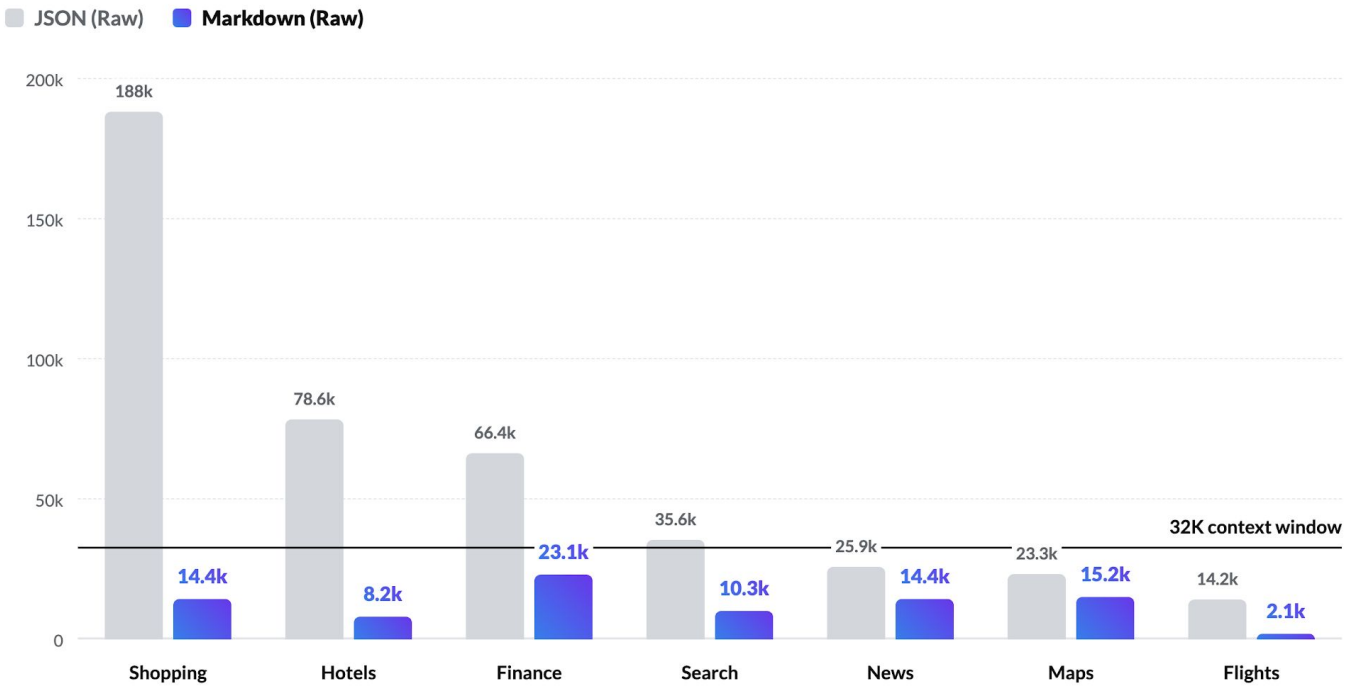
326,949

tokens, almost 10 times the window

MARKDOWN, RAW

71,244

tokens, 4.6 times smaller



Token count of the full response, estimated with tiktoken (cl100k_base). Each model had a 32,768-token context window.

SELECT FIELDS, USE MARKDOWN

Send the Model Only What It Needs

Select fields

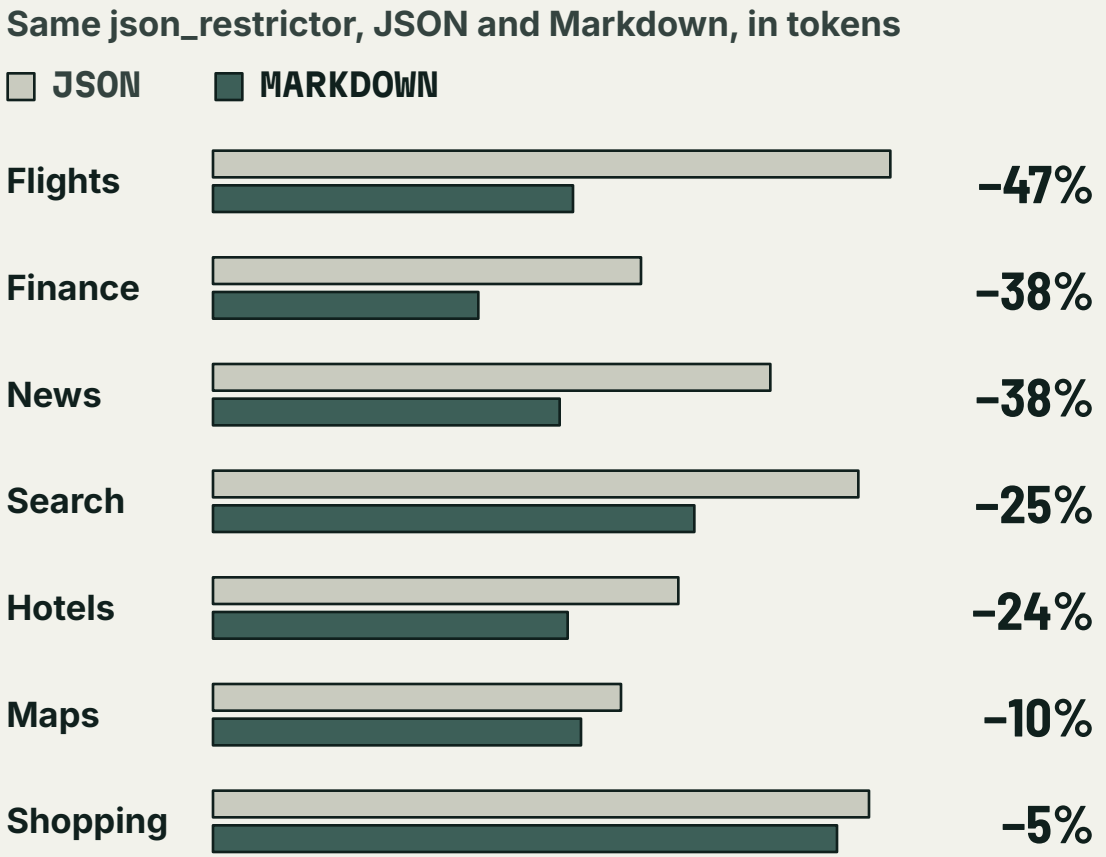
`json_restrictor`

SerpApi trims each response to the fields the model needs, about 500 to 900 tokens per search.

Use Markdown

`output=md`

The same fields take 27% fewer tokens, so the restrictor can keep more results.



LIVE DEMO

The App, with SerpApi as the Search Tool

PROMPT 1

Who is the current CEO of Apple?

PROMPT 2

Hotels in Barcelona, December 9 to 12, for 2 adults

PROMPT 3

Compare AAPL and TSLA today

github.com/serpapi/local-llm-web-search

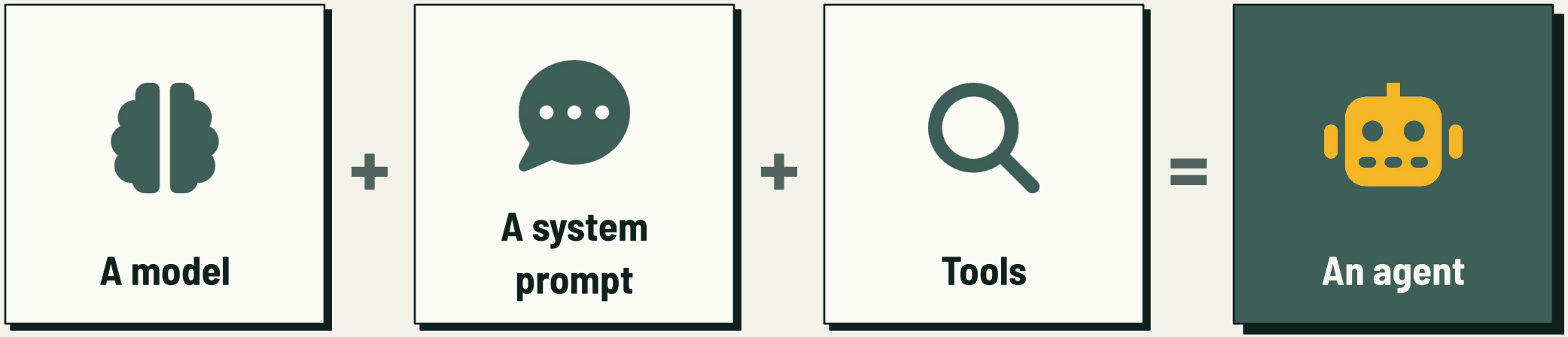
02

From Tools to MCP, Skills and CLI

Four ways to extend an agent, in Python

AGENT BASICS

An Agent Is a Model, a System Prompt and Tools



Answers from its training data

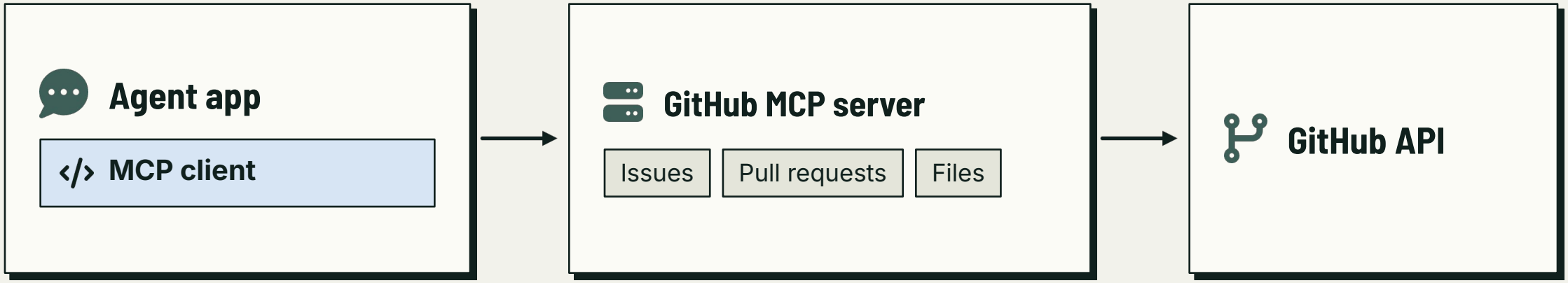
What it is for and how it behaves

Allow the agent to interact with the world

Gets a current answer

MCP

One Protocol for Reusable Integrations



Discover available tools and their schemas.

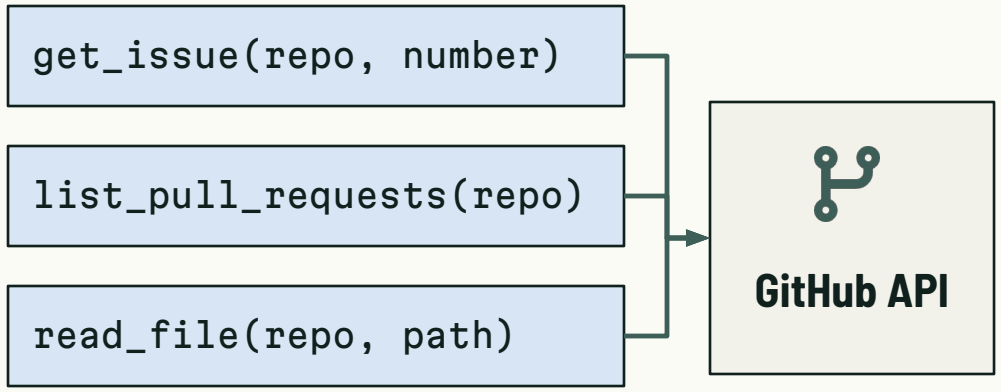
Call them through a common protocol.

MCP connects capabilities. Your agent runtime still runs the loop.

MCP · THE GITHUB EXAMPLE

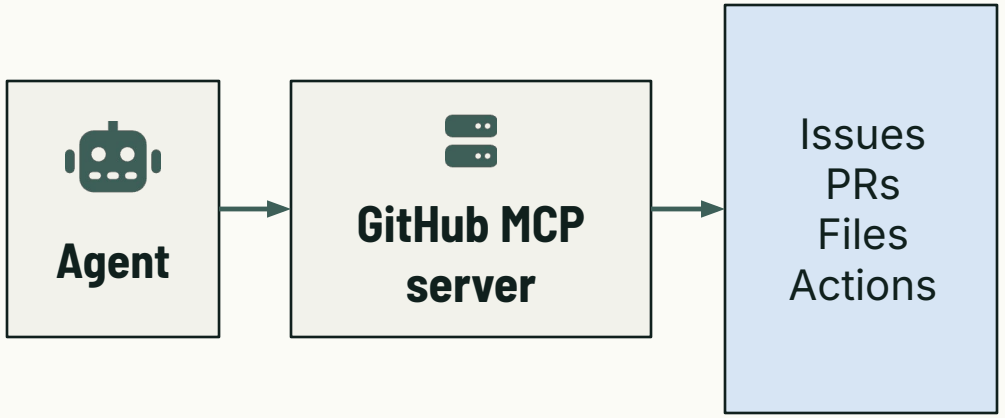
Build the Wrappers, or Reuse an Integration

Custom tools



You design and maintain each exposed operation.

MCP



Reuse the server's tool definitions and implementation.

Broad coverage can require many wrappers. Narrow agents may need only a few.

MCP · WHEN TO USE

Use MCP to Reuse a Service Integration

An existing server fits

GitHub, search or another service already exposes useful tools.

Several clients need access

Reuse the integration across compatible agent apps.

You need broader coverage

Avoid maintaining a large set of API wrappers.

You delegate

Tool implementation and server behavior

You retain control

Tool selection, credentials, permissions and approval gates

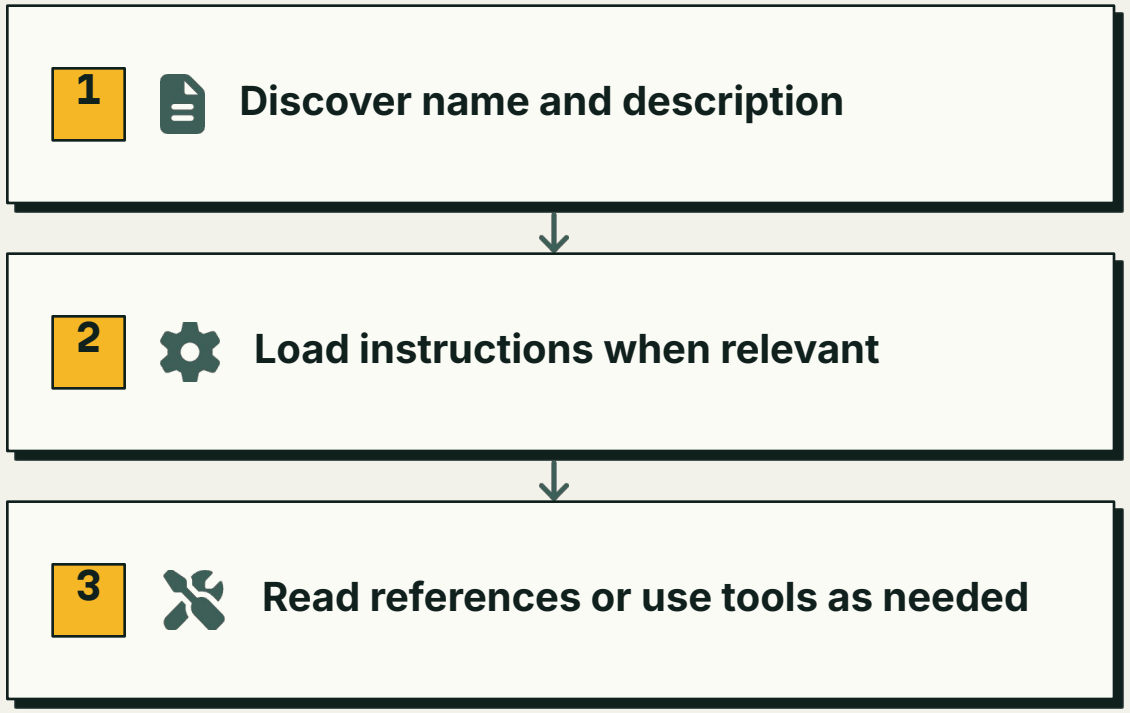
SKILLS

Shared Instructions for a Type of Work

A SKILL IS A FOLDER

```
meeting-actions/
├── SKILL.md
├── references/
└── scripts/
```

SKILL.md holds the metadata and instructions.
References and scripts are optional.



Example: turn rough meeting notes into your team's handoff format.

SKILLS · WHEN TO USE

Use Skills to Teach a Repeatable Workflow

Team conventions

Meeting handoffs, review checklists and writing rules.

Task knowledge

Which search engine, fields and sources to use.

Reusable guidance

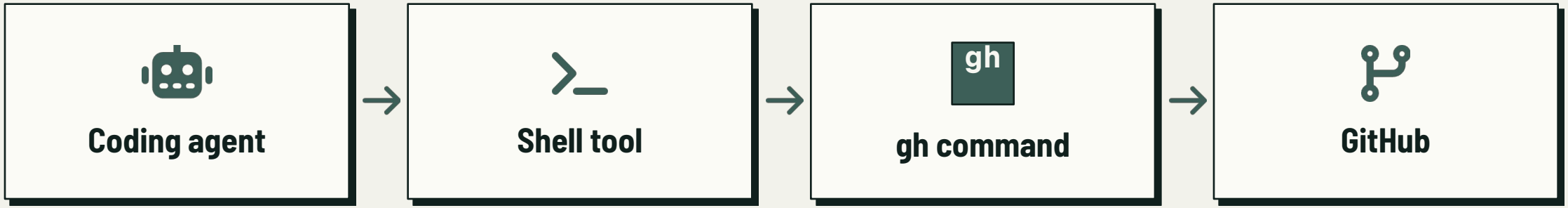
Share one procedure across compatible agents.

→ **Skills guide how the model works.**

→ **Use code and tool validation for rules that must be enforced.**

CLI

Give a Coding Agent Familiar Commands




```
gh issue list --repo org/repo --json number,title
```

Read issues as structured JSON.

The shell is a tool. The CLI is the program it runs.

CLI · WHEN TO USE

Use CLI Tools Inside Developer Workflows

An existing terminal setup

The agent already has a shell and the right tools.

Established commands

Use gh, git, test runners and build tools.

Scriptable output

Request JSON and check exit codes.

A wrong command can change the wrong thing

Wrong repository, flags or credentials can cause unwanted edits, merges or deletions.

- Start read-only
- Restrict permissions
- Review high-impact writes

PART 3 • TOGETHER

Tool Calls, MCP, CLI or Skills?

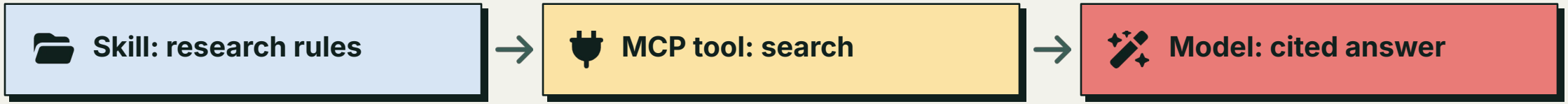
APPROACH	RUNS THE LOOP	GOOD FIT WHEN
Tool calls	Your code	You control every payload, as on a local model
MCP	MCP Client	The client can discover and use the server defined tool
CLI	The agent, through a shell	The agent already has a terminal, like a coding agent
Skills	The agent, with instructions	You want to teach a workflow, not only add a tool

Your pick for a local model on a laptop? For a team on Claude or ChatGPT?

PUTTING IT TOGETHER

Choose for the Constraint, Combine as Needed

NEED	REACH FOR	EXAMPLE
Exact inputs and business rules	**Custom tools**	Validate a refund request
A reusable service integration	**MCP**	GitHub issues and pull requests
Existing developer commands	**CLI**	gh, git and test runners
A shared task procedure	**Skills**	Research or meeting handoffs



These are complementary layers, not mutually exclusive choices.

DEV CHALLENGES • HACKTOBERFEST 2026

Hacktoberfest Open-Source AI Challenge

"Build something with open-source AI at its core."

Week 3 • Slow and Steady • Oct 19 to 25

Week 4 • Oct 26 to 31

Run your own model

Clone the app, load an open model and run the Benchmark page.

Add an ability

A tool for another search engine, or the loop in your language.

Best Use of SerpApi

A partner category for live web search, fresh RAG results or news tracking.

THE CHALLENGES
dev.to/challenges/hf26

THE APP
github.com/serpapi/local-llm-web-search

Thanks for Watching





RESOURCES

Session materials
`serpapi.com/hacktoberfest`

How to Fit More Web Search Data into a Local LLM
`serpapi.com/blog/local-llm-web-search-markdown-output`

How to Connect Your Local LLM with Web Search Data
`serpapi.com/blog/how-to-connect-your-local-llm-with-web-search-data`

Introducing SerpApi's New Markdown Output
`serpapi.com/blog/introducing-serpapi-markdown-output`

The app's source code
`github.com/serpapi/local-llm-web-search`